COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA

B.Sc DEGREE EXAMINATION

COLLEGE: Science and Technology **DEPARTMENT:** Mathematics
SESSION: 2023/2024 **SEMESTER:** Alpha
COURSE CODE: MAT213 **CREDIT UNIT:** 3
COURSE TITLE: Ordinary Differential Equations I **TIME**: 3 Hours

INSTRUCTION: *Attempt any* ***four*** *questions.*

QUESTION ONE

a) Identify and list the steps to solve a differential equation of the form $y' + p(x)y = g(x)$ using the integrating factor method. **(9 marks)**

b) Solve the initial value problem: $y' - 2y = te^{-2t}$, $y(0) = 1$. **(8.5 marks)**

QUESTION TWO

a) Given an ordinary differential equation $xy + (2x^2 + 3y^2 - 20)y' = 0$, you are required to:

i) find the general solution, **(8 marks)**

ii) find the solution that satisfies the initial condition $y(0) = 2$. **(2 marks)**

b) Solve the initial value problem $y' = \frac{xy^3}{\sqrt{1+x^2}}$, $y(0) = 1$. **(7.5 marks)**

QUESTION THREE

a) A radioactive with a mass of 50gms. was found to have a mass of 40gms. after 30 years. Find its half-life. **(8 marks)**

b) Find the general solution of $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 4e^x$ using the method of variation of parameters. **(9.5 marks)**.

QUESTION FOUR

a) Find the general solution the equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 1 = -e^x$, using the method of undetermined coefficient. **(9.5 marks)**

b) A rod of temperature 100^0C is kept in a surrounding of temperature 20^0C. If the temperature of the rod was found to be 80^0C after 10 minutes, find the temperature $T(t)$ of the rod. **(8 marks)**.

QUESTION FIVE

(a) Define the Laplace Transform of $f(t)$. That is, $L\{f(t)\}$. **(2 marks)**

(b) Show that:

i) $L\{k\} = \frac{k}{s}$ **(3 marks)**

ii) $L\{f'(t)\} = sL\{f(t)\} - f(0)$ **(3 marks)**

(c) State and prove two properties of Laplace Transform. **(5.5 marks)**

(d) Classify the following second order PDEs into hyperbolic, parabolic or elliptic:

i) $\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2}$ **(2 marks)**

ii) $\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial x} + 4\frac{\partial^2 u}{\partial y^2} - \frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial x} = 0$ **(2 marks)**

QUESTION SIX

(a) Evaluate the following:

i) $L^{-1}\left\{\frac{3s+1}{s^2 - s - 6}\right\}$ **(3 marks)**

ii) $L^{-1}\left\{\frac{9s-8}{s^2 - 2s}\right\}$ **(3 marks)**

(b) Find the Laplace transforms of the following functions:

i) $3t^3e^{-4t}$ **(2 marks)**

ii) $t\sin 3t$ **(2 marks)**

iii) $e^{3t}\sin 2t$ **(2 marks)**

(c) Solve the IVP: $y' - 2y = 4$; $y(0) = 1$ by the inverse Laplace transform method **(5.5 marks)**